

Comparitive Performance Report for tentacle-ec-4+2-16k-balance-reads-off- run2 vs direct-reads-pr2-ec-4+2-16k-balance-off vs direct-reads-pr2-ec-4+2-16k-balance-on- run2 vs direct-reads-alexpoc-ec-4+2-16k-ec- balance-reads-on vs tentacle-3-way-replica-balance-reads-on

Table of contents

- Comparison summary for tentacle-ec-4+2-16k-balance-reads-off-run2 vs direct-reads-pr2-ec-4+2-16k-balance-off vs direct-reads-pr2-ec-4+2-16k-balance-on-run2 vs direct-reads-alexpoc-ec-4+2-16k-ec-balance-reads-on vs tentacle-3-way-replica-balance-reads-on
- Response Curves
 - Sequential Read
 - Random Read
 - Random Read/Write
- Configuration yaml files
 - results

Comparison summary for tentacle-ec-4+2-16k-balance-reads-off-run2 vs direct-reads-pr2-ec-4+2-16k-balance-off vs direct-reads-pr2-ec-4+2-16k-balance-on-run2 vs direct-reads-alexpoc-ec-4+2-16k-ec-balance-reads-on vs tentacle-3-way-replica-balance-reads-on

Sequential

Read	tentacle_ec_4+2-16k	direct_16k	pr2-ec-4+2-16k	balance-reads-off-run2	balance-off	balance-on-run2	alexpoc-ec-4+2-16k-ec-balance-reads-on	tentacle-3-way-replica-balance-reads-on	%change	direct_16k
4K	89458@4.3m	89308@4.3	-0%	151063@2.5	69%	278599@1.1	211%	261426@1.5	192%	
8K	90112@4.3m	90051@4.3	-0%	148569@2.6	65%	296498@1.3	229%	252900@1.5	181%	
16K	88672@4.3m	88958@4.3	0%	144892@2.6	63%	274597@1.4	210%	219887@1.7	148%	
32K	73365@5.2m	74468@5.1	2%	56357@6.8	-23%	136779@2.8	86%	192102@2.0	162%	
64K	3182@7.9ms	3296@7.6	4%	2368@10.6	-26%	4385@4.8	38%	9299@2.7	192%	
128K	4861@10.3m	5394@9.3	11%	4457@11.3	-8%	7430@4.5	53%	13091@3.8	169%	
256K	6711@15.0m	7239@9.3	8%	7482@13.4	11%	10882@7.7	62%	17215@5.8	157%	
384K	6865@29.3m	8059@25.0	17%	8750@23.0	27%	12153@5.2	77%	16934@11.9	147%	
512K	7275@36.9m	8712@30.8	20%	10333@13.0	42%	13295@10.1	83%	14250@18.8	96%	
768K	7488@26.9m	8867@45.4	18%	11093@9.1	48%	14458@3.5	93%	16851@11.9	125%	
1024K	8047@16.7m	9069@44.4	13%	12541@21.4	56%	15636@17.2	94%	15398@17.4	91%	
2048K	8274@48.7m	9018@44.7	9%	13579@9.9	64%	15937@8.4	93%	14321@28.1	73%	
4096K	7953@16.9m	8673@61.9	9%	12707@10.5	60%	15027@17.8	89%	14110@57.1	77%	

Random

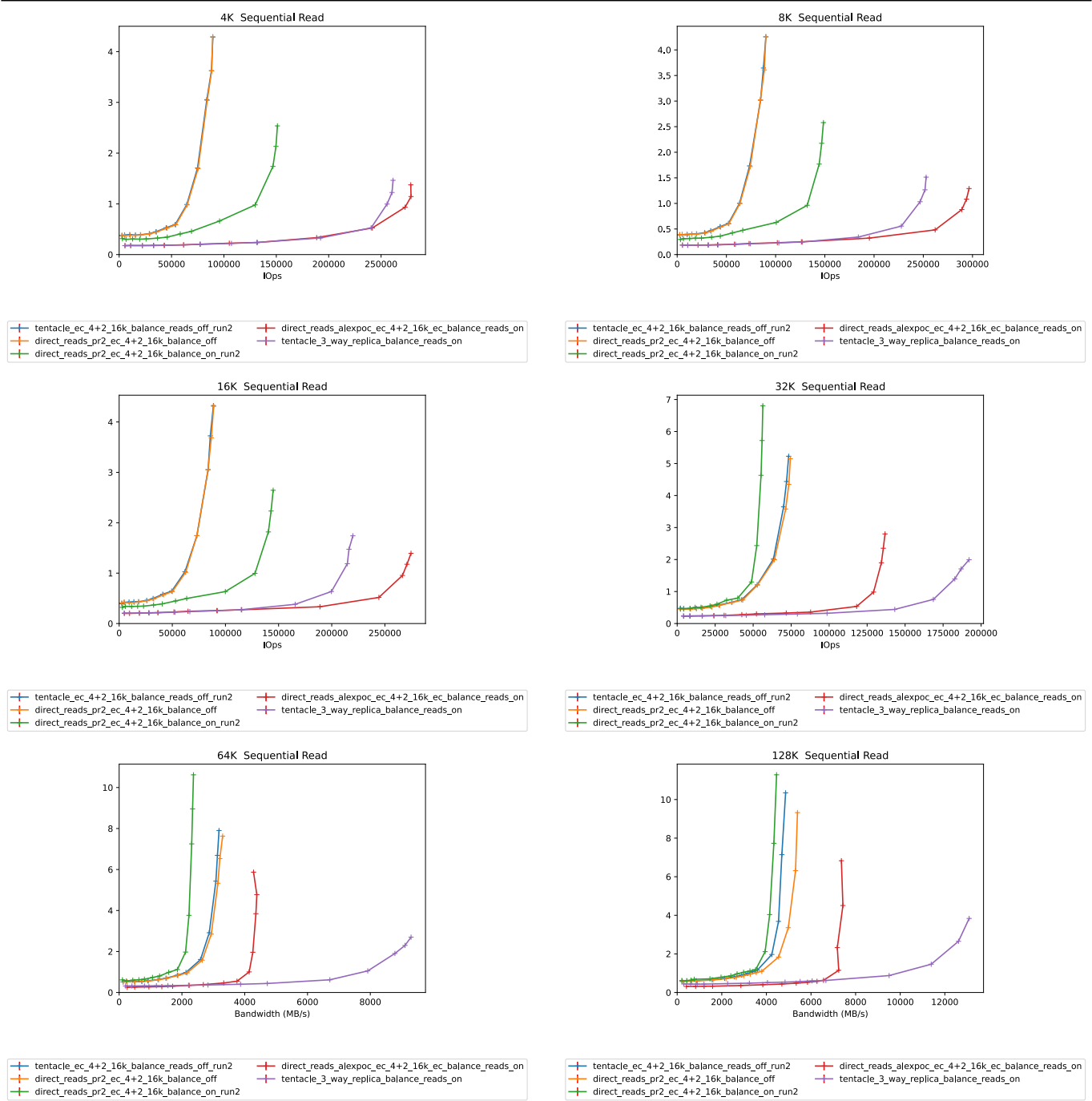
Read	tentacle_ec_4+2-16k	direct_16k	pr2-ec-4+2-16k	balance-reads-off-run2	balance-off	balance-on-run2	alexpoc-ec-4+2-16k-ec-balance-reads-on	tentacle-3-way-replica-balance-reads-on	%change	direct_16k
4K	109637@3.5m	112826@3.4	3%	147506@2.6	35%	227436@1.7	107%	239094@1.1	118%	
8K	105499@3.6m	108129@3.5	2%	145356@2.6	38%	213634@1.2	102%	225342@1.7	114%	
16K	100733@3.8m	101958@3.8	1%	139453@2.7	38%	197284@1.6	96%	203290@1.9	102%	
32K	68357@4.7m	71338@4.5	4%	59527@6.4	-13%	97798@2.6	43%	182074@1.8	166%	
64K	2780@6.0ms	2930@7.2	5%	2350@10.7	-15%	3258@7.7	17%	9412@2.7	239%	
128K	4613@7.3ms	4975@10.1	8%	4216@11.9	-9%	5962@8.4	29%	13212@2.5	186%	
256K	6111@5.5ms	6840@4.9	12%	6922@12.1	13%	9672@10.4	58%	14106@7.1	131%	
384K	6789@3.7ms	7644@13.2	13%	8486@23.7	25%	11572@17.4	70%	16696@3.8	146%	
512K	6933@19.4m	8193@32.8	18%	9894@27.1	43%	13001@20.6	88%	15960@4.2	130%	
768K	7598@53.0m	8678@46.4	14%	10690@18.8	41%	14051@28.7	85%	16308@6.2	115%	
1024K	7620@8.8ms	8938@30.0	17%	11855@34.0	56%	15352@17.5	101%	16095@8.3	111%	
2048K	8553@47.1m	9101@29.5	6%	13221@10.1	55%	16058@8.3	88%	14222@18.8	66%	
4096K	8481@31.6m	8731@30.7	3%	13557@19.8	60%	14800@9.0	75%	13727@58.7	62%	

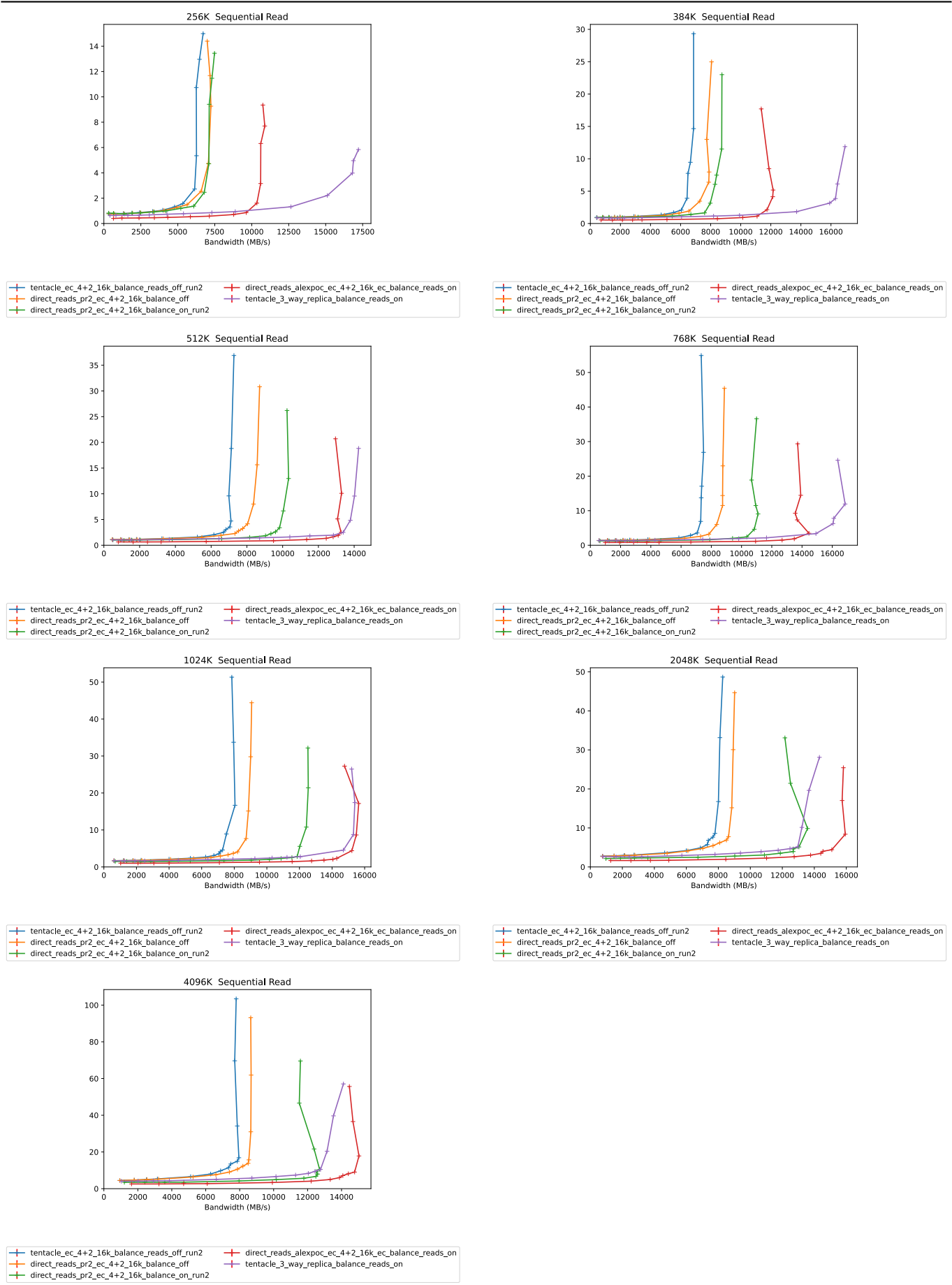
Comparison summary for tentacle-ec-4+2-16k-balance-reads-off-run2 vs direct-reads-pr2-ec-4+2-16k-balance-off vs direct-reads-pr2-ec-4+2-16k-balance-on-run2 vs direct-reads-alexpoc-ec-4+2-16k-ec-balance-reads-on vs tentacle-3-way-replica-balance-reads-on

Random									
Read/Write	tentacle_ec_direct_16k	direct_16k	pr2_on	pr2_off	pr2_on	pr2_off	pr2_on	pr2_off	pr2_on
4K_70/30	46104@8.3m	49029@6.5	6%	53288@6.0	16%	64072@5.0	39%	123422@3.1	168%
16K_70/30	42227@9.1m	46327@8.3	10%	49888@7.7	18%	58460@6.6	38%	106175@3.6	151%
64K_30/70	1303@19.3m	1394@18.0	7%	1305@19.3	0%	1454@14.4	12%	2947@2.8	126%
64K_70/30	1741@12.0m	1822@11.5	5%	1617@15.5	-7%	1994@12.6	15%	4942@5.1	184%

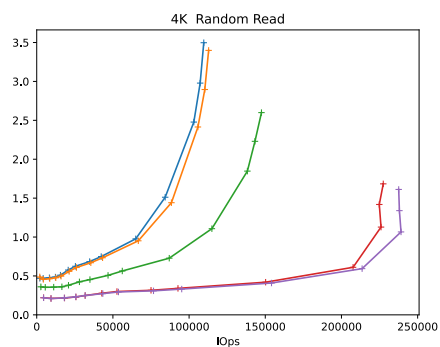
Response Curves

Sequential Read

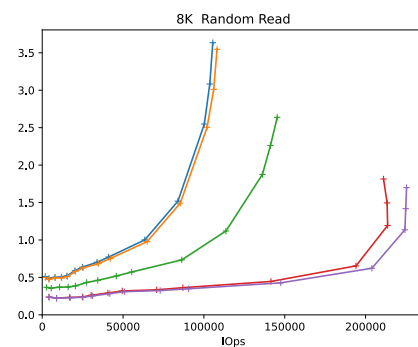




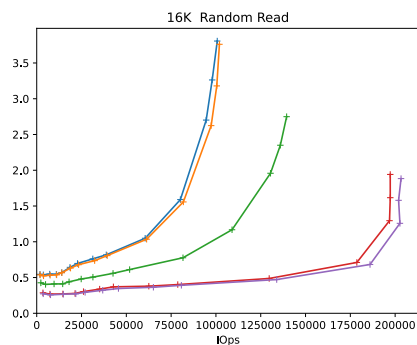
Random Read



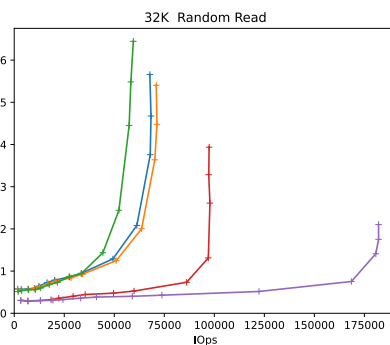
+ tentacle_ec_4+2_16k_balance_reads_off_run2 + direct_reads_alexpc_ec_4+2_16k_ec_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_off + tentacle_3_way_replica_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_on_run2



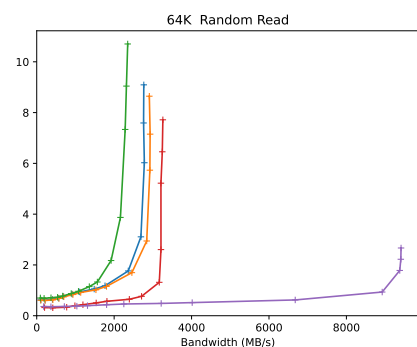
+ tentacle_ec_4+2_16k_balance_reads_off_run2 + direct_reads_alexpc_ec_4+2_16k_ec_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_off + tentacle_3_way_replica_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_on_run2



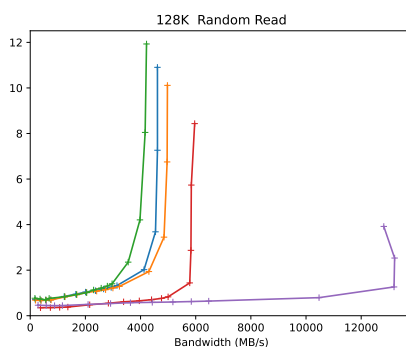
+ tentacle_ec_4+2_16k_balance_reads_off_run2 + direct_reads_alexpc_ec_4+2_16k_ec_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_off + tentacle_3_way_replica_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_on_run2



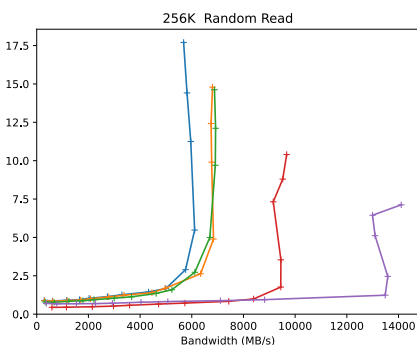
+ tentacle_ec_4+2_16k_balance_reads_off_run2 + direct_reads_alexpc_ec_4+2_16k_ec_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_off + tentacle_3_way_replica_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_on_run2



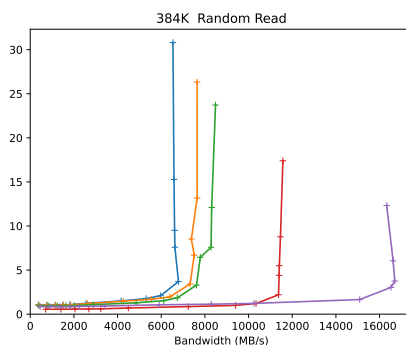
+ tentacle_ec_4+2_16k_balance_reads_off_run2 + direct_reads_alexpc_ec_4+2_16k_ec_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_off + tentacle_3_way_replica_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_on_run2



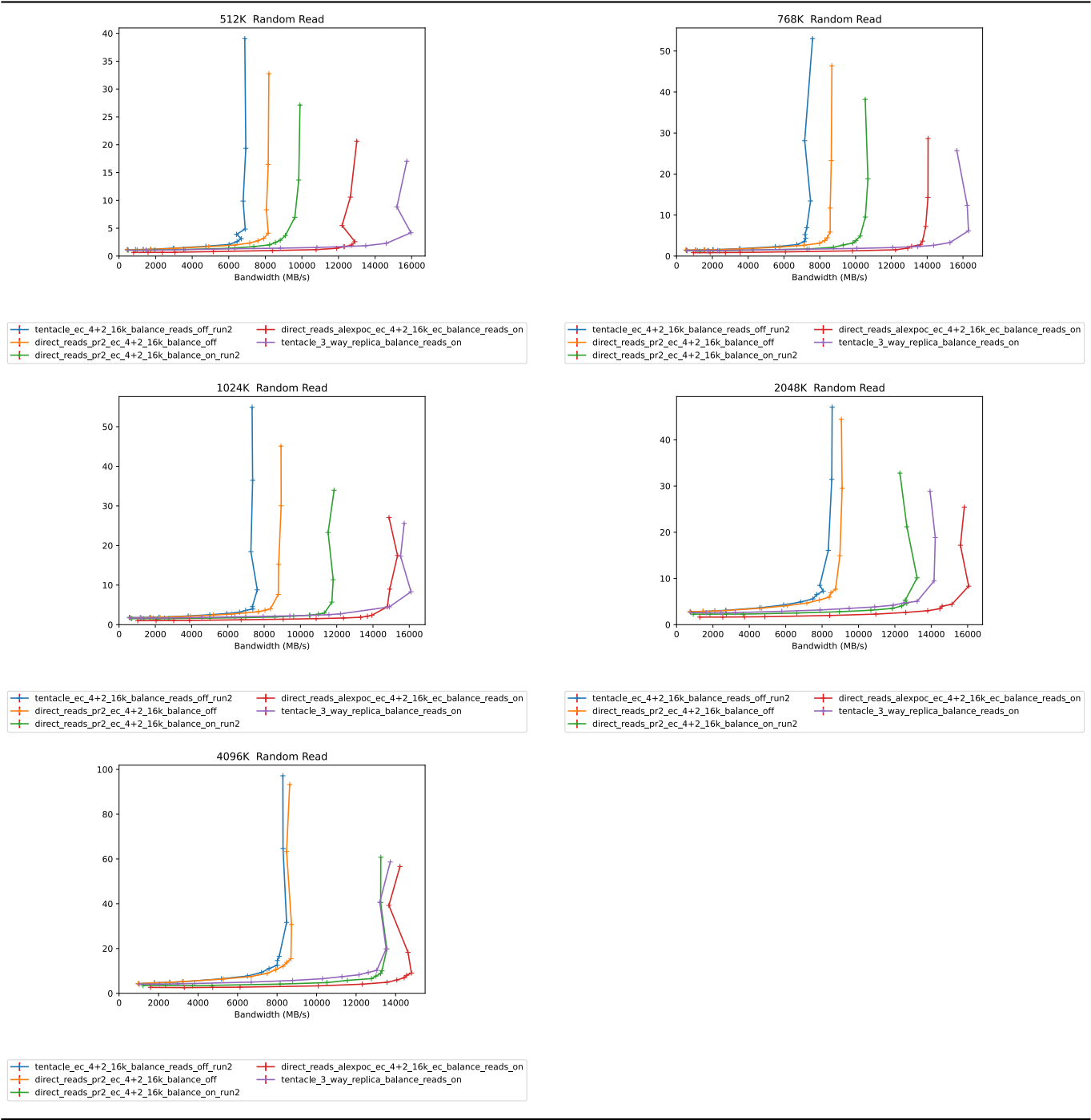
+ tentacle_ec_4+2_16k_balance_reads_off_run2 + direct_reads_alexpc_ec_4+2_16k_ec_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_off + tentacle_3_way_replica_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_on_run2



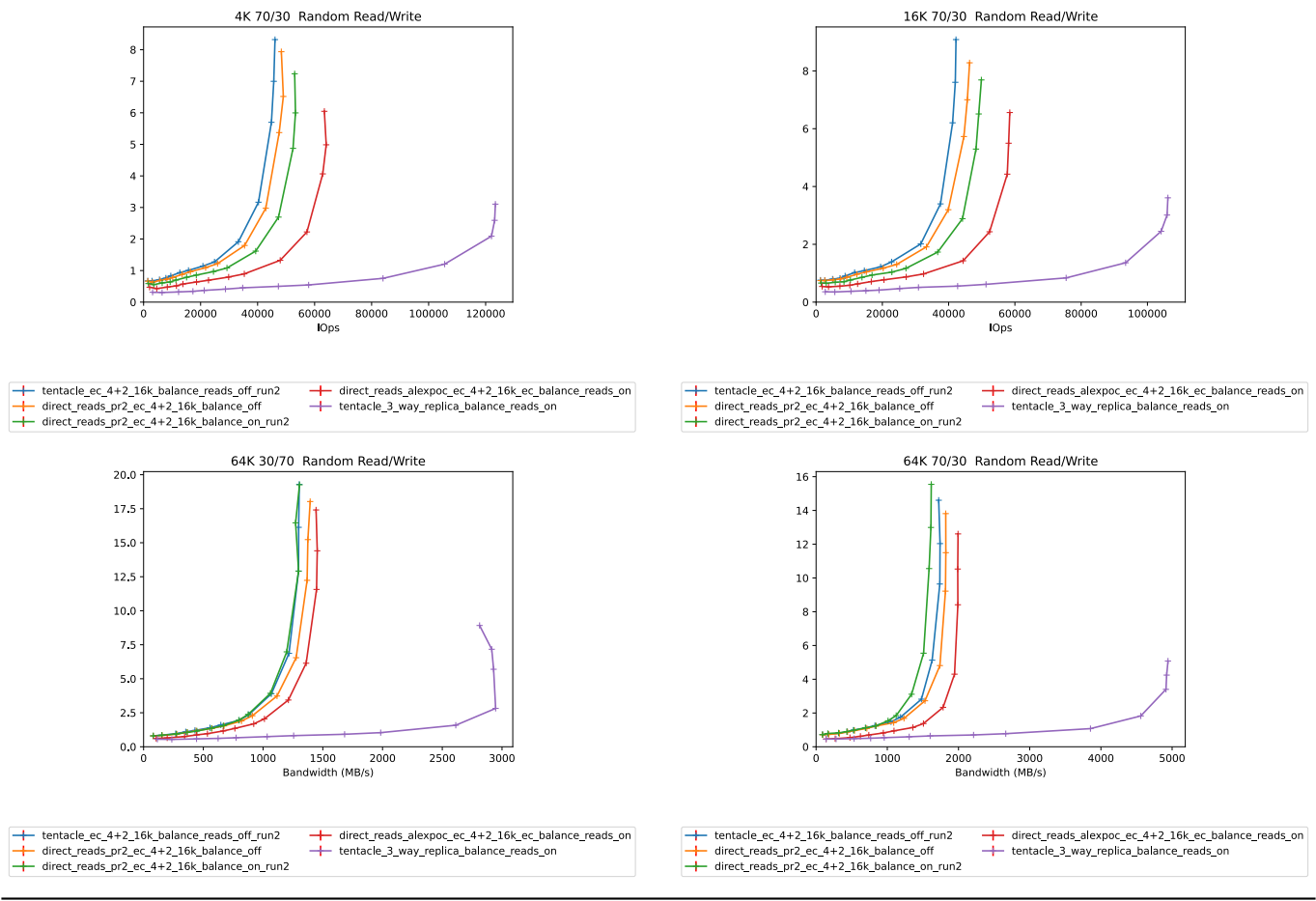
+ tentacle_ec_4+2_16k_balance_reads_off_run2 + direct_reads_alexpc_ec_4+2_16k_ec_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_off + tentacle_3_way_replica_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_on_run2



+ tentacle_ec_4+2_16k_balance_reads_off_run2 + direct_reads_alexpc_ec_4+2_16k_ec_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_off + tentacle_3_way_replica_balance_reads_on
 + direct_reads_pr2_ec_4+2_16k_balance_on_run2



Random Read/Write



Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384

16kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 16384
- rwmixread: 70
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

16krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 16384
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

16ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 16384
- total_iodepth:
- 1
- 2
- 4

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1

```


- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

```

4kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 4096
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:

```

```

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30

```

```

    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
64kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 65536
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
64ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1

```

```

  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph_mgr_cmd: /usr/bin/ceph-mgr
  ceph_mon_cmd: /usr/bin/ceph-mon
  ceph_osd_cmd: /usr/bin/ceph-osd
  ceph_run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```

    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
osds_per_node: 6
pdsh_ssh_args: -a -x -l%u %h
rados_cmd: /usr/bin/rados
rbd_cmd: /usr/bin/rbd
tmp_dir: /tmp/cbt
use_existing: true
user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -FO --rawtoo --sep ";" -f {collectl_dir}

```